

Johannes Schimunek

Postdoctoral Researcher in Machine Learning

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Profile

Machine-learning researcher with a Ph.D. in Machine Learning / Chemoinformatics and a publication record at the interface of machine learning, chemistry, and biology. I develop data-efficient models for molecular and biological AI, with particular emphasis on few-shot learning, molecular property prediction, generative sequence modeling, and rigorous evaluation in scientific workflows. My work combines strong mathematical foundations with interdisciplinary collaboration in drug-discovery and life-science settings.

Research strengths: few-shot and low-data learning; molecular property prediction / QSAR modeling; generative modeling for biological and chemical sequences; structured reasoning and diagnostic evaluation.

Research Experience

Postdoctoral Researcher

2025–present

Johannes Kepler University Linz & LIT AI Lab

- Co-developed *MolecularIQ* (ICLR 2026), a benchmark for chemical reasoning with symbolic verification and controlled analysis of invariances, consistency, and failure modes.
- Contributed to a collaborative project on structured constraint repair in collaborative data settings (ESWC 2026), outside my main research focus.
- Continue working on reliable and data-efficient machine learning for molecular and biological AI systems, with emphasis on structured reasoning, evaluation, and collaboration across technical and application domains.

Scientific Researcher

2019–2025

Johannes Kepler University Linz & LIT AI Lab

- Designed and implemented end-to-end machine-learning pipelines for few-shot bioactivity prediction, autoregressive activity prediction, and foundation-model style learning across small molecules, proteins, and DNA.
- Developed *MHNfs* (JCIM 2025) and *Context-enriched molecule representations improve few-shot drug discovery* (ICLR 2023) for low-data molecular property prediction and few-shot drug discovery.
- Contributed to *Bio-xLSTM* (ICLR 2025), covering generative modeling, representation learning, and in-context learning for biological and chemical sequences.
- Worked on machine-learning-guided directed protein optimization and other collaborative scientific-discovery projects linking machine learning with chemistry and biology, including industry collaboration in drug-discovery settings.
- Managed end-to-end experiments including data preparation, model development, hyperparameter tuning, ablations, error analysis, and clear communication of results to collaborators from machine learning and life-science backgrounds.

Mathematician, Statistician & Data Scientist

2016–2017

SSI Software Services GmbH, Munich

- Developed algorithms for predictive maintenance, pattern recognition, and process stabilization in industrial settings.
- Translated applied operational problems into analytical workflows in collaboration with technical stakeholders.

Account Manager, B2B Software Solutions

2015–2016

SSI Software Services GmbH, Munich

Supported startup-oriented software projects with client communication, problem scoping, and solution coordination.

Education

Dr. rer. nat. in Machine Learning / Chemoinformatics

2019–2025

Johannes Kepler University Linz

Thesis: *Machine learning for low-data drug discovery*. Supervisors: Günter Klambauer and Sepp Hochreiter. External examiner: Andrea Volkamer.

Second state examination in Mathematics and Physics teaching

2017–2019

Max-Planck-Gymnasium München

School-based teacher training (*Referendariat*) with practical teaching experience in mathematics and physics.

First state examination in Mathematics and Physics teaching

2011–2017

University of Augsburg

Graduate-level training in mathematics, physics, and pedagogy.

B.Sc. in Mathematics

2014–2015

University of Augsburg

Bachelor thesis on the statistical analysis of ski races.

Teaching, Supervision, and Service

- Authorized examiner for Master's theses at Johannes Kepler University Linz since 2026; supervised more than ten bachelor's and master's theses.
- University teaching in *AI for Life Sciences*, *Deep Learning and Neural Networks*, *Deep Learning: Basic Techniques*, *Python II*, *Hands on AI*, and *Programming in Python* as exercise lecturer, guest lecturer, or teaching support.
- Organizer and host of a weekly Drug Discovery Journal Club since 2020.
- Reviewer for ICLR, NeurIPS, ICML, *Journal of Chemical Information and Modeling*, *Journal of Medicinal Chemistry*, and *Transactions on Machine Learning Research*.
- Organizing-committee roles for the ELLIS ML4Molecules workshop (2024, 2025) and related community events in machine learning for life sciences.

Selected Distinctions and Invited Talks

- ELLIS ML4Molecules best paper award (2024) for *Bio-xLSTM: Generative modeling, representation and in-context learning of biological and chemical sequences*.
- AI for Good Spotlight (ICLR 2024 DMLR workshop) for *Autoregressive activity prediction for low-data drug discovery*.
- Top Reviewer for NeurIPS 2024.
- Invited talks at the AI for Good Global Summit (2024) and the ELLIS Health Symposium on Machine Learning for Drug Discovery (2024).

Publications

Note: * denotes equal contribution.

Peer-reviewed publications

1. Bartmann*, Schimunek*, Ielanskyi, Seidl, Klambauer, Luukkonen. **MolecularIQ: Characterizing chemical reasoning capabilities through symbolic verification on molecular graphs**. *International Conference on Learning Representations (ICLR)*, 2026. Status: Accepted.
2. Vázquez, Innerebner, Prock, Klambauer, Lex, Schimunek, Polleres. **Structure is the Signal: Graph Encodings and GNNs for Constraint Repair in Collaborative Knowledge Graphs**. *European Semantic Web Conference (ESWC)*, 2026. Status: Accepted.
3. Schimunek, Klambauer. **Few-shot learning in drug discovery**. In *Applied Artificial Intelligence for Drug Discovery*. Springer Nature, 2025. Status: Accepted.
4. Schmidinger, Schneckenreiter, Seidl, Schimunek, Hoedt, Brandstetter, Mayr, Luukkonen, Hochreiter, Klambauer. **Bio-xLSTM: Generative modeling, representation and in-context learning of biological and chemical sequences**. *International Conference on Learning Representations (ICLR)*, 2025. Status: Accepted.
5. Schimunek, Luukkonen, Klambauer. **MHNfs: Prompting in-context bioactivity predictions for low-data drug discovery**. *Journal of Chemical Information and Modeling (JCIM)*, 2025. Status: Accepted.
6. Schimunek, Friedrich, Kuhn, Klambauer. **Autoregressive activity prediction for low-data drug discovery**. *ICLR Workshop on practical ML for limited/low resource settings and Workshop on Data-centric Machine Learning Research (DMLR)*, 2024. Status: Accepted and published.
7. Schimunek, Seidl, Friedrich, Kuhn, Rippmann, Hochreiter, Klambauer. **Context-enriched molecule representations improve few-shot drug discovery**. *International Conference on Learning Representations (ICLR)*, 2023. Status: Accepted and published.
8. Schimunek, Seidl, Von Delft, Walsh, Horvath, Brooks III, et al. **A community effort in SARS-CoV-2 drug discovery**. *Molecular Informatics*, 2024. Status: Accepted and published.
9. Schimunek, Friedrich, Kuhn, Rippmann, Hochreiter, Klambauer. **A generalized framework for embedding-based few-shot learning methods in drug discovery**. *ELLIS Machine Learning for Molecule Discovery Workshop (ELLIS ML4Mol)*, 2021. Status: Accepted and published.
10. Schimunek, Friedrich, Kuhn, Rippmann, Hochreiter, Klambauer. **Comparative assessment of interpretability methods of deep activity models for hERG**. *International Workshop on (Q)SAR in Environmental and Health Sciences*, 2021. Status: Accepted and published.

Under review

1. Ielanskyi, Wang, Scott, Rieber, Merrett, Schimunek, Mayr, McDowell, Klambauer, Bowen. **Machine-learning assisted optimization of cytosine and adenine base editors derived from ADATs**. *Nucleic Acids Research*, 2026. Status: Under review.